数理方程

第三周
20260313

内容：ODE · 分离变量法

ODE　一阶ODE

1) 可分 separable

$$\frac{dy}{dt}=\frac{g(t)}{f(y)} \Rightarrow f(y)dy=g(t)dt$$
$$\Rightarrow F(y)=G(t)+C$$

例：

Logistic eq. $\frac{dy}{dt}=ay\left(1-\frac{y}{N}\right) \xrightarrow{x=y/N} \frac{dx}{dt}=ax(1-x)$

$$\Rightarrow \int\frac{dx}{x(1-x)}=\int a\,dt \Rightarrow \ln\frac{x}{1-x}=at+\text{const} \Rightarrow x(t)=\frac{1}{1+Ce^{-at}}$$

C可用$x(0)$表示

$$=\frac{x(0)}{x(0)+[1-x(0)]e^{-at}}$$

例：$\frac{dy}{dt}+a(t)y=0, \Rightarrow \frac{dy}{y}=-a(t)dt$

$$\Rightarrow \ln|y|=-\int a(t)dt+\text{const} \Rightarrow \pm y=|y|=\text{const}\cdot e^{-\int a(t)dt}$$

$$y(t)=Ce^{-\int a(t)dt}$$

$\frac{dy}{dt}+6ty=0,\ y(0)=5 \Rightarrow a(t)=6t$

$$y(t)=Ce^{-\int 6t\,dt}=Ce^{-3t^2}=5e^{-3t^2}$$

2) 一阶线性ODE

$$\frac{dy}{dt}+a(t)y=b(t),\quad y(t_0)=y_0$$

$$\mu(t)\frac{dy}{dt} + a(t)\mu(t)y = b(t)\mu(t)$$

⇓ 选取 μ 使 $\frac{d\mu}{dt} = a(t)\mu(t)$ ⇓ $\mu(t) = e^{\int a(t)dt}$

$$\frac{d}{dt}(\mu(t)y(t)) = b(t)\mu(t)$$

$$\Rightarrow \mu(t)y(t) = \int b(t)\mu(t)dt + C$$

$$\Rightarrow \mu(t)y(t) - \mu(t_0)y_0 = \int_{t_0}^{t} b(s)\mu(s)ds$$

$$\Rightarrow y(t) = \frac{1}{\mu(t)}\left[\mu(t_0)y_0 + \int_{t_0}^{t} b(s)\mu(s)ds\right]$$

3) 恰当方程 $M(x,y)dx + N(x,y)dy = 0$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ $\Rightarrow d\phi = Mdx + Ndy$

$\phi_x = M$, $\phi_y = N$

$\phi_{xy} = \phi_{yx}$

$\Rightarrow \phi(x,y) = C$

$\phi_x dx + \phi_y dy = 0$

$y^3dx + 3xy^2dy = 0$

$\phi_x = y^3$, $\phi_y = 3xy^2$ $\Rightarrow$ $\phi = xy^3$

$\Rightarrow xy^3 = C$

乘 y^2

$ydx + 3xdy = 0$ ← 不满足 $M_y = N_x$

二阶 ODE

a, b, c 实常数

$L(y_1+y_2) = L(y_1) + L(y_2)$ 线性

1) 常系数线性方程

$y(t)$, $ay'' + by' + cy = L(y) = 0$

试探 $y = e^{rt}$, $L(e^{rt}) = a(e^{rt})'' + b(e^{rt})' + ce^{rt}$

$= (ar^2 + br + c)e^{rt} = 0$

$\Rightarrow ar^2 + br + c = 0$ 特征方程

$\Rightarrow r_{1,2} = \frac{1}{2a}(-b \pm \sqrt{b^2-4ac})$

① 两实根 $r_1 \neq r_2$ $\quad y(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}$

② 两复根 $\alpha \pm i\beta$

$y(t) = C_1 e^{(\alpha+i\beta)t} + C_2 e^{(\alpha-i\beta)t}$

$= C_1 e^{\alpha t}(\cos\beta t + i\sin\beta t) + C_2 e^{\alpha t}(\cos\beta t - i\sin\beta t)$

$= (C_1+C_2)e^{\alpha t}\cos\beta t + i(C_1-C_2)e^{\alpha t}\sin\beta t$

$= \tilde{C}_1 e^{\alpha t}\cos\beta t + \tilde{C}_2 e^{\alpha t}\sin\beta t$

例: $y''+a^2y=0 \Rightarrow r^2+a^2=0 \Rightarrow r_{1,2}=\pm ia \Rightarrow y = C_1\cos at + C_2\sin at$

③ $r_1=r_2=-\frac{b}{2a}=r \Rightarrow y(t)=(C_1+C_2t)e^{rt}$

$L(e^{\alpha t}) = (a\alpha^2+b\alpha+c)e^{\alpha t} = a(\alpha-r)^2 e^{\alpha t}$ (r: 重根)

$\Downarrow \frac{d}{d\alpha}$

$L(\frac{d}{d\alpha}e^{\alpha t}) = 2a(\alpha-r)e^{\alpha t} + a(\alpha-r)^2 t e^{\alpha t}$

$\Vert$

$L(te^{\alpha t})$

取 $\alpha = r \Rightarrow L(te^{rt}) = 0$

推广至高阶: $a_n y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0$

$\Rightarrow a_n r^n + \cdots + a_1 r + a_0 = 0 \Rightarrow$ 根 $r_1, r_2 \cdots$

无重根: $y = \sum_{j=1}^{n} C_j e^{r_j t}$ (复根项可化为三角函数)

一个 m 重根 $r \Rightarrow (C_1 + C_2 t + \cdots C_m t^{m-1})e^{rt} + \sum_{\text{其他根}} (\cdots)$

2) 非齐次方程 $y''+p(t)y'+q(t)y=f(t)$

常数变易法，先解 $y''+py'+qy=0 \Rightarrow C_1y_1(t)+C_2y_2(t)$

特解 $y(t)=C_1(t)y_1(t)+C_2(t)y_2(t)$，求 $C_1(t), C_2(t)$，略

齐次化原理：若 $w(t;\tau)$ 满足 $\begin{cases}\frac{d^2w}{dt^2}+p(t)\frac{dw}{dt}+q(t)w=0, & t\geq\tau \\ w|_{t=\tau}=0, \quad \frac{dw}{dt}|_{t=\tau}=f(\tau)\end{cases}$

$\Rightarrow y(t)=\int_0^t w(t;\tau)d\tau$ 满足

$$\begin{cases}y''+py'+qy=f \\ y(0)=0, \quad y'(0)=0\end{cases}$$

例：$y''+a^2y=f(t)$, $y(0)=y'(0)=0$ （$a>0$）

$\Rightarrow w(t;\tau)$ $\begin{cases}w''+a^2w=0, & t>\tau \\ w|_{t=\tau}=0, \quad \frac{dw}{dt}|_{t=\tau}=f(\tau)\end{cases}$

$\Rightarrow w(t;\tau)=A(\tau)\sin at+B(\tau)\cos at$

$=A\sin a(t-\tau)$ ← $w|_{t=\tau}=0$

$=\frac{1}{a}\sin a(t-\tau)f(\tau)$ ← $\frac{dw}{dt}|_{t=\tau}=f(\tau)$

$\Rightarrow y(t)=\int_0^t \frac{1}{a}\sin a(t-\tau)f(\tau)d\tau$

比常数变易法更简便、直观

$f(t)=\sin at \Rightarrow y(t)=\int_0^t \frac{1}{a}\sin a(t-\tau)\sin a\tau\, d\tau=\cdots$ 类推.

问题，$y''-a^2y=f(t)$, $y(0)=y'(0)=0 \Rightarrow y(t)=\int_0^t \frac{1}{a}\sinh a(t-\tau)f(\tau)d\tau$

若初始条件 $y(0)=A\neq 0$, $y'(0)=B\neq 0$?

$$\Rightarrow y(t)=\underbrace{C_1\sin at+C_2\cos at}_{\text{齐次方程通解}}+\underbrace{\int_0^t \frac{1}{a}\sin a(t-\tau)f(\tau)\,d\tau}_{\text{非齐次方程特解}}$$

$$\Downarrow$$

$$C_2=A,\ C_1=\frac{B}{a}$$

$L(y_1)=0$, $L(y_2)=f$ $\Rightarrow L(y_1+y_2)=f$

3) 欧拉方程

$$x^2\frac{d^2y}{dx^2}+\alpha x\frac{dy}{dx}+\beta y=0$$

$\Rightarrow t=\ln x$ $\quad x=e^t$, $\quad \frac{dx}{dt}=e^t=x$,

$$\frac{d}{dt}=\frac{dx}{dt}\frac{d}{dx}=x\frac{d}{dx}$$

$$\frac{d^2}{dt^2}=x\frac{d}{dx}\left(x\frac{d}{dx}\right)=x^2\frac{d^2}{dx^2}+x\frac{d}{dx}$$

$$\Downarrow$$

$$x^2\frac{d^2}{dx^2}=\frac{d}{dt^2}-\frac{d}{dt}$$

$\frac{d}{dt}$

$$\Rightarrow \frac{d^2y}{dt^2}-\frac{dy}{dt}+\alpha\frac{dy}{dt}+\beta y=0$$

$$\Rightarrow \frac{d^2y}{dt^2}+(\alpha-1)\frac{dy}{dt}+\beta y=0 \qquad r^2+(\alpha-1)r+\beta=0$$

① $r_1\neq r_2 \Rightarrow y(t)=C_1e^{r_1t}+C_2e^{r_2t}=C_1x^{r_1}+C_2x^{r_2}$

② $r_{1,2}=\lambda\pm i\mu \Rightarrow y=e^{\lambda t}(C_1\cos\mu t+C_2\sin\mu t)$

$$=x^{\lambda}\left[C_1\cos(\mu\ln x)+C_2\sin(\mu\ln x)\right]$$

③ $r_1=r_2=r \Rightarrow y=(C_1+C_2t)e^{rt}=(C_1+C_2\ln x)x^r$

也可不引入t,

直接以 $y=x^r$ 代入原方程 $\Rightarrow \underbrace{[r(r-1)+\alpha r+\beta]}_{=0\ \text{特征方程}}x^r=0$

另外，有些方程可猜特解形式，加速计算.

例：$x''+cx'+\omega^2 x=f\cos\alpha t$ ① 阻尼受迫振动.

可先解 $y''+cy'+\omega^2 y=fe^{i\alpha t}$ ②

若②有解 $y(t)$，方程两边取实部可得 $x(t)=\mathrm{Re}\, y(t)$ 是①的解.

对②，可猜特解为 $y(t)=Ae^{i\alpha t}$（待定），代入②

$$\Rightarrow (-\alpha^2+ic\alpha+\omega^2)Ae^{i\alpha t}=fe^{i\alpha t}$$

$$\Rightarrow A=\frac{f}{\omega^2-\alpha^2+ic\alpha}$$

$$=|A|e^{-i\phi},$$

$$|A|=\frac{f}{\sqrt{(\omega^2-\alpha^2)^2+(c\alpha)^2}}$$

$$\tan\phi=\frac{c\alpha}{\omega^2-\alpha^2}$$

$$\Rightarrow y(t)=|A|e^{i(\alpha t-\phi)}$$

$$\Rightarrow x(t)=\mathrm{Re}\, y(t)=|A|\cos(\alpha t-\phi)$$

本例也展示了复数对计算的简化.

ODE本征值问题 = 阶ODE, 例: $\frac{d^2y}{dx^2} + \lambda y = 0$ (λ: 常数)

给定一点值: $y(0) = y'(0) = 0 \Rightarrow y(x) = 0$

给定两点的值 $= 0$: $y(0) = y(\pi) = 0 \overset{?}{\Rightarrow} y(x) = 0$

以 $\lambda > 0$ 为例: $y(x) = A\sin(\sqrt{\lambda}x) + B\cos(\sqrt{\lambda}x)$

$y(0) = 0 \Rightarrow B = 0$, $y(\pi) = 0 \Rightarrow A\sin\sqrt{\lambda}\pi = 0$

$A \neq 0 \Rightarrow \sin\sqrt{\lambda}\pi = 0 \Rightarrow \sqrt{\lambda}\pi = n\pi$, $n = 1, 2, \cdots$

$\lambda_n = n^2$ 本征值

$y(x) = A\sin nx$ 本征函数

$\lambda =$ 本征值 $\Rightarrow y(x) = A\sin nx$

$\lambda \neq$ 本征值 $\Rightarrow y(x) = 0$

矩阵 本征值

$MX = \lambda X$. $X \neq 0 \Rightarrow \det(M - \lambda \mathbb{1}) = 0$

一阶ODE, 例: $-i\frac{dy}{dx} = \lambda y$, $y(0) = y(2\pi)$

$\Rightarrow y = Ae^{i\lambda x}$, $Ce^{i\lambda\cdot 0} = Ce^{i\lambda\cdot 2\pi} \Rightarrow e^{2\pi\lambda i} = 1 \Rightarrow \lambda_n = n \in \mathbb{Z}$ (整数)

$y_n(x) = e^{inx}$

分离变量法

PDE → ODE + ODE

PDE → ODE + ODE + ODE

傅里叶, Fourier 1768-1830

热传导方程



$u_t = k u_{xx}$ $\quad u|_{x=0} = u|_{x=l} = 0$ $\quad u|_{t=0} = \phi(x)$

1) 分离变量 $u(x,t) = X(x)T(t)$

↑ 满足 PDE + 边界条件，~~初始条件~~ 暂时忽略

$$PDE \Rightarrow X(x)T'(t) = kX''(x)T(t)$$

$$\Rightarrow \frac{T'(t)}{kT(t)} = \frac{X''(x)}{X(x)} = -\lambda$$

↑不依赖 x ↑不依赖 t

$$\Rightarrow \boxed{X''(x) + \lambda X(x) = 0}, \quad T'(t) + \lambda k T(t) = 0$$

$u|_{x=0} = X(0)T(t) = 0 \Rightarrow \boxed{X(0) = 0}$

$u|_{x=l} = X(l)T(t) = 0 \Rightarrow \boxed{X(l) = 0}$

ODE 本征值问题

$u_t = k u_{xx} + f(x,t) \Rightarrow T'X = kX''T + f(x,t)$

非齐次 PDE

$$\Rightarrow \frac{T'}{kT} = \frac{X''}{X} + \frac{f(x,t)}{kXT} \Rightarrow ?$$

$u|_{x=0} = \mu(t) = X(0)T(t) \Rightarrow T(t) = \frac{\mu(t)}{X(0)}$?

非齐次边界条件

齐次 PDE + 齐次边界条件 才可直接分离变量

2) 本征值问题 $\begin{cases} X''(x)+\lambda X(x)=0 \\ X(0)=X(l)=0 \end{cases}$

也可不假定λ为实数，但计算会发现λ只有实数解

↓

Sturm-Liouville 理论（后面学）$\Rightarrow$ 本征值λ为实数

a) $\lambda=0$, $X(x)=Ax+B \xrightarrow{X(0)=X(l)=0} A=B=0$

b) $\lambda<0$, $X''-|\lambda|X=0 \Rightarrow X=Ae^{\sqrt{|\lambda|}x}+Be^{-\sqrt{|\lambda|}x}$

$X(0)=A+B=0$, $X(l)=Ae^{\sqrt{|\lambda|}l}+Be^{-\sqrt{|\lambda|}l}=0$

$\Rightarrow A=B=0$

c) $\lambda>0$, $X(x)=A\sin\sqrt{\lambda}x+B\cos\sqrt{\lambda}x$

$X(0)=0 \Rightarrow B=0 \Rightarrow X=A\sin\sqrt{\lambda}x$ $A=1$

$X(l)=0 \Rightarrow \sqrt{\lambda}l=n\pi$, $n=1,2,\cdots \Rightarrow$ 本征值$\lambda_n=\left(\frac{n\pi}{l}\right)^2$

本征函数$X_n=\sin\frac{n\pi}{l}x$

3) 级数型解 $u_n=X_n(x)T_n(t)$

$T_n'(t)+\lambda_n k T_n(t)=0 \Rightarrow T_n(t)=Be^{-\lambda_n k t}=Be^{-k\left(\frac{n\pi}{l}\right)^2 t}$（常数↓）

$\Rightarrow u_n(x,t)=X_n(x)T_n(t)=B\sin\frac{n\pi x}{l}e^{-k\left(\frac{n\pi}{l}\right)^2 t}$

↑ 问题已解决

对特殊初始条件 $u|_{t=0}=B\sin\frac{n\pi}{l}x$，正弦型初始条件

两个正弦之和：

$$u|_{t=0}=7\sin\frac{3\pi x}{l}+5\sin\frac{6\pi x}{l}\Rightarrow u(x,t)=7\sin\frac{3\pi x}{l}e^{-k(\frac{3\pi}{l})^2t}+5\sin\frac{6\pi x}{l}e^{-k(\frac{6\pi}{l})^2t}$$

多个正弦之和：

$$u|_{t=0}=\sum_{n=1}^{M}B_n\sin\frac{n\pi}{l}x\Rightarrow u(x,t)=\sum_{n=1}^{M}B_n\sin\frac{n\pi x}{l}e^{-k(\frac{n\pi}{l})^2t}$$

↑ 非齐线型

4) 完整解.

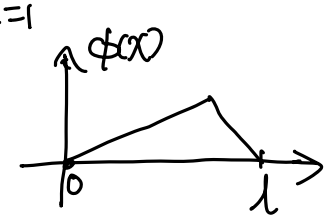
如果 $u|_{t=0}=\phi(x)$ 是一个一般的函数？ 不能写成有限个正弦之和

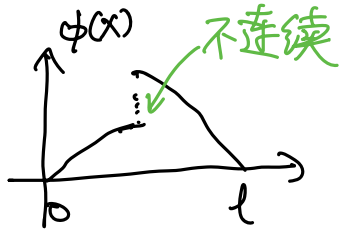
Fourier：$[0,l]$ 上"任意"函数 $\phi(x)$ 可展开为 Fourier 级数：

$$\phi(x)=\sum_{n=1}^{\infty}B_n\sin\frac{n\pi x}{l}=\sum_{n=1}^{\infty}B_nX_n(x)$$

$$\Downarrow$$

$$u(x,t)=\sum_{n=1}^{\infty}B_n\sin\frac{n\pi x}{l}e^{-k(\frac{n\pi}{l})^2t}$$





若 $\phi(x)$ 可展开，

$B_n=?$ ⇐ 正交性 $\int_0^l X_n(x)X_m(x)\,dx=0,\quad n\neq m$

$$\phi(x)=\sum_{m=1}^{\infty}B_mX_m(x)\Longrightarrow\int_0^l\phi(x)X_n(x)\,dx=B_n\int_0^l[X_n(x)]^2dx$$

$=\|X_n\|^2$ 模方

$$\Rightarrow B_n=\frac{1}{\|X_n\|^2}\int_0^l\phi(x)X_n(x)\,dx$$

$$=\frac{2}{l}\int_0^l\phi(x)\sin\frac{n\pi x}{l}dx$$

$$\|X_n\|^2=\int_0^l\left(\sin\frac{n\pi x}{l}\right)^2dx=\frac{l}{2}$$

回到热传导：



例：$u|_{t=0} = \phi(x) = 1$，$\Rightarrow B_n = \frac{2}{n\pi}[1-(-1)^n] = \begin{cases} \frac{4}{n\pi} & (n \text{ odd}) \\ 0 & (n \text{ even}) \end{cases}$

$$\Rightarrow u(x,t) = \sum_{n=1}^{\infty} B_n \sin\frac{n\pi x}{l} e^{-k(\frac{n\pi}{l})^2 t}$$

$$= \sum_{n \text{ odd}} \frac{4}{n\pi} \sin\frac{n\pi x}{l} e^{-k(\frac{n\pi}{l})^2 t}$$

靠前的项（n较小）构画轮廓，靠后的项补充细节.

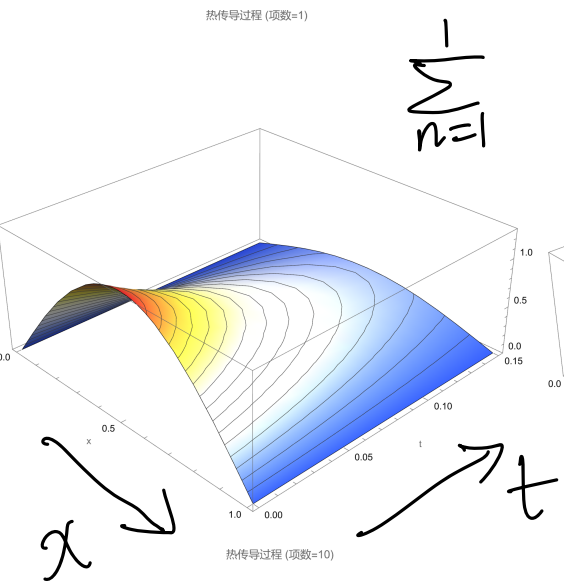
见下页图像.

图像展示

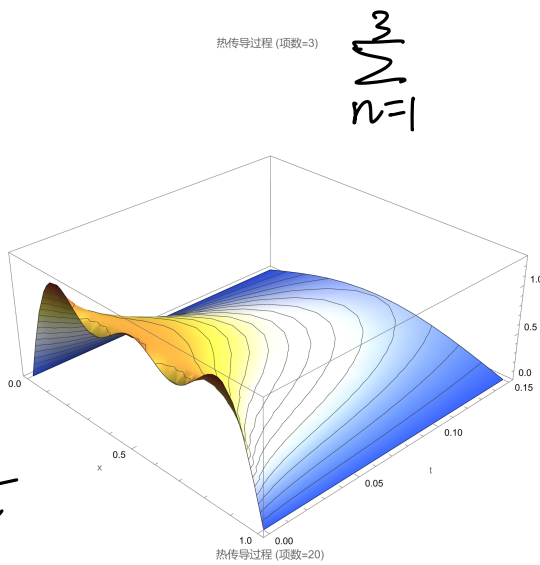
$u_t = u_{xx}, \quad 0<x<1, \ t>0$

$u|_{x=0} = u|_{x=1} = 0, \quad u|_{t=0} = 1$

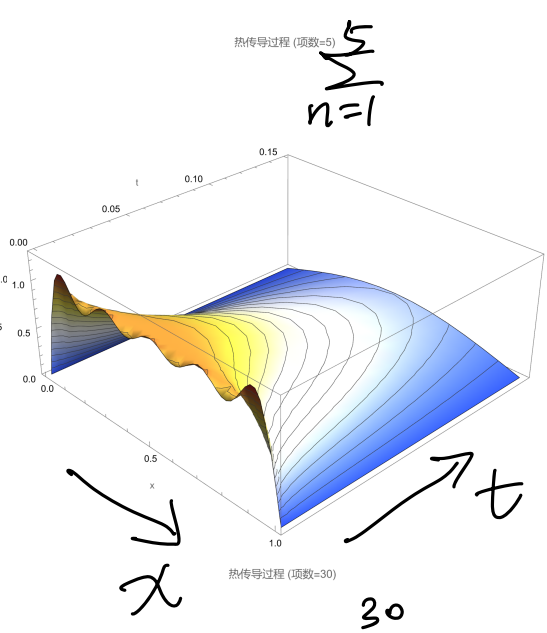
取 $k=1, l=1$



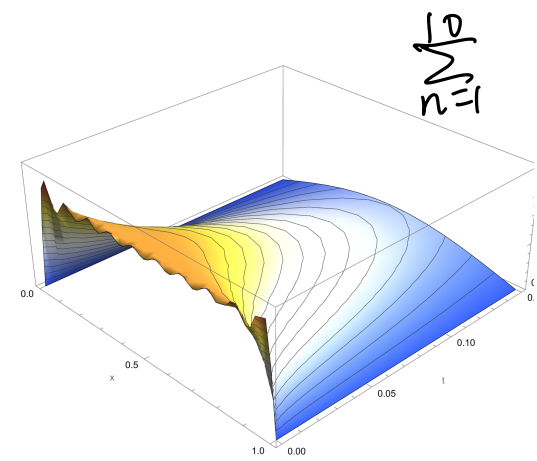
$\sum_{n=1}^{1}$



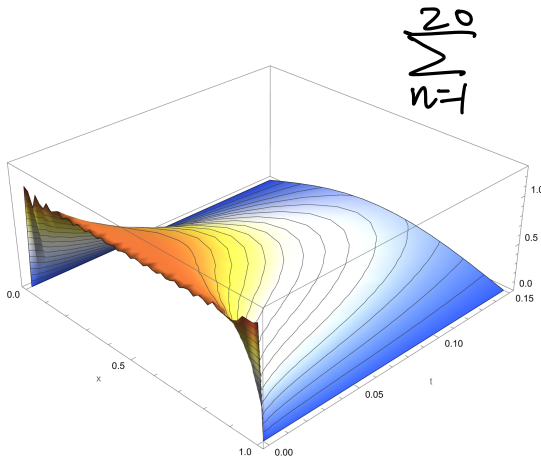
$\sum_{n=1}^{3}$



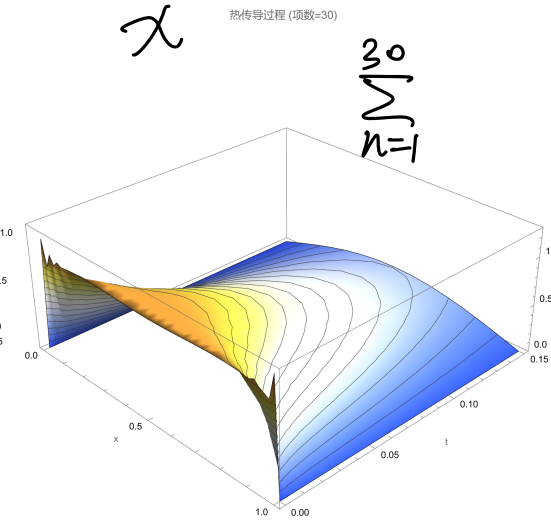
$\sum_{n=1}^{5}$



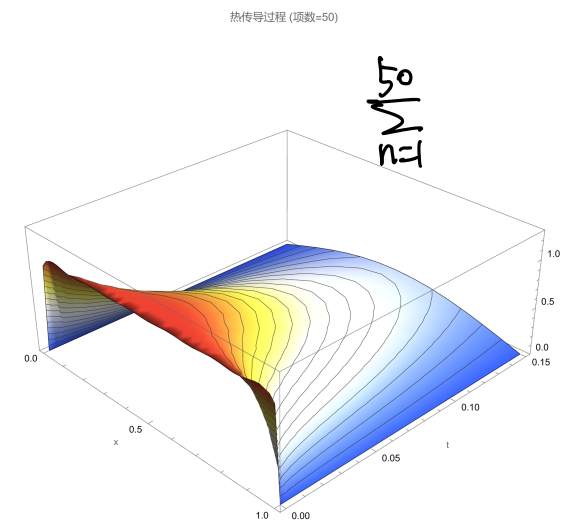
$\sum_{n=1}^{10}$



$\sum_{n=1}^{20}$



$\sum_{n=1}^{30}$



$\sum_{n=1}^{50}$

$$u(x,t) = \sum_{n\ odd} \frac{4}{n\pi} \sin\frac{n\pi x}{l} e^{-k\left(\frac{n\pi}{l}\right)^2 t}$$

对于不太小的 t $\left(t > \frac{l^2}{k\pi^2}\right)$，n越大，随t衰减越快

少数几项即可给出很好近似.

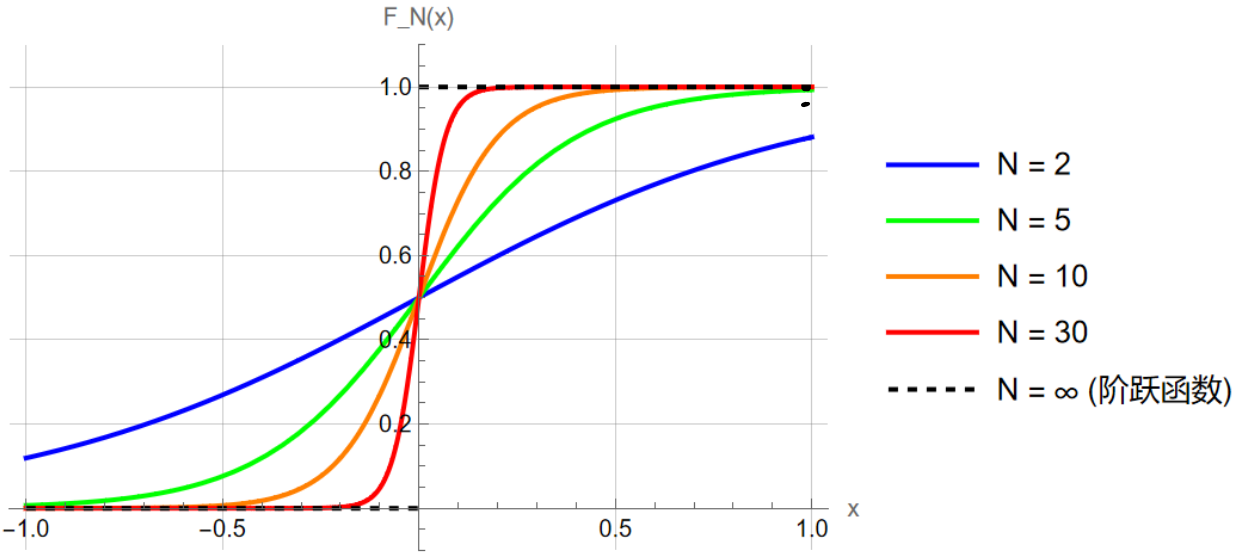
为什么性质优良(例如,连续,可微..)的正弦函数求和能表示性质不佳(例如不连续)的函数?

无穷和 $\sum_{n=1}^{\infty} f_n$ 并不继承部分和 $\sum_{n=1}^{N} f_n$ 的性质(例如连续性).

$\sum_{n=1}^{\infty} f_n$: 部分和的 $N \to \infty$ 极限

例: $f_n(x) = \begin{cases} F_1(x) & (n=1) \\ F_n(x) - F_{n-1}(x) & \end{cases}$ $\Rightarrow F_n(x) = \frac{1}{1+e^{-nx}}$

$\sum_{n=1}^{N} f_n(x) = F_N(x)$ $\begin{cases} N\text{有限}: & \text{连续、可微} \\ N=\infty: & \text{不连续} \end{cases}$



关于收敛性概念：

函数 $\phi(x)$ 可展开为 Fourier 级数 $\sum_{n=1}^{\infty} B_n X_n(x)$

即 $\sum_{n=1}^{M} B_n X_n(x) \xrightarrow{M\to\infty} \phi(x)$

（收敛至）

"收敛"的几种不同含义：

① 逐点收敛 (pointwise)

② 一致收敛 (uniformly)

③ 七均方收敛 (mean square)

$$\int_0^l \left[\phi(x) - \sum_{n=1}^{M} B_n X_n(x)\right]^2 dx \xrightarrow{M\to\infty} 0$$

① ② ③ 对 $\phi(x)$ 性质各有要求

Fourier 级数收敛对 $\phi(x)$ 性质要求比较宽松. 与 Taylor 展开比

详情参见：E. Stein, R. Shakarchi 《傅里叶分析》(有中译本)

W. Strauss 《偏微分方程导论》,

R. Haberman. 《实用偏微分方程》

傅里叶正弦级数 $[0,l]$. $f(x)=\sum_{n=1}^{\infty} B_n \sin\frac{n\pi x}{l}$, $B_n=\frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}dx$

傅里叶余弦级数 $[0,l]$. $f(x)=\sum_{n=0}^{\infty} C_n \cos\frac{n\pi x}{l}$

$$C_n=\begin{cases}\frac{1}{l}\int_0^l f(x)dx & n=0\\ \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx & n\geq 1\end{cases}$$

傅里叶级数 $[-l,l]$. $f(x)=a_0+\sum_{n=1}^{\infty} a_n\cos\frac{n\pi x}{l}+\sum_{n=1}^{\infty} b_n\sin\frac{n\pi x}{l}$.

$a_0=\frac{1}{2l}\int_{-l}^{l} f(x)dx$, $a_n=\frac{1}{l}\int_{-l}^{l} f(x)\cos\frac{n\pi x}{l}dx$, $b_n=\frac{1}{l}\int_{-l}^{l} f(x)\sin\frac{n\pi x}{l}dx$

$n\geq 1$

三种级数都是 Sturm-Liouville 理论的特例. 区别在于 $\{X_n(x)\}$ 不同，对应于不同的本征值问题. 课程后续有 S-L 理论详情.

波动方程解法相似，请自行阅读、计算

波动方程：$u_{tt} - a^2 u_{xx} = 0, \quad 0<x<l, \ t>0$

$u|_{x=0} = u|_{x=l} = 0$



$u|_{t=0} = \phi(x), \quad u_t|_{t=0} = \psi(x), \quad 0 \le x \le l$

1) 分离变量 $u = X(x)T(t) \Rightarrow XT'' = a^2 X'' T$

$$\Rightarrow \frac{T''(t)}{a^2 T(t)} = \frac{X''(x)}{X(x)} = -\lambda \Rightarrow \begin{cases} X''(x) + \lambda X(x) = 0 \\ T''(t) + \lambda a^2 T(t) = 0 \end{cases}$$

$u|_{x=0} = X(0)T(t) = 0 \Rightarrow X(0) = 0$

$\cdots \Rightarrow X(l) = 0$

2) 本征值问题 $\lambda_n = \left(\frac{n\pi}{l}\right)^2, \ n = 1, 2 \cdots \quad X_n = \sin\frac{n\pi x}{l}$

3) $T_n'' + \lambda_n a^2 T_n(t) = 0, \quad T_n'' + \left(\frac{n\pi a}{l}\right)^2 T_n = 0$

$$T_n(t) = C_n \sin\frac{n\pi a}{l} t + D_n \cos\frac{n\pi a}{l} t$$

$$u_n = X_n(x) T_n(t) = \left(C_n \sin\frac{n\pi a}{l} t + D_n \cos\frac{n\pi a}{l} t\right) \sin\frac{n\pi}{l} x$$

4) $$u(x,t) = \sum_{n=1}^{\infty} u_n(x,t) = \sum_{n=1}^{\infty} \left(C_n \sin\frac{n\pi a}{l} t + D_n \cos\frac{n\pi a}{l} t\right) \sin\frac{n\pi}{l} x$$

$$u|_{t=0} = \phi(x) = \sum_{n=1}^{\infty} D_n \sin\frac{n\pi x}{l} \Rightarrow D_n = \frac{2}{l}\int_0^l \phi(x) \sin\frac{n\pi}{l} x \, dx$$

$$u_t|_{t=0} = \psi(x) = \sum_{n=1}^{\infty} C_n \frac{n\pi a}{l} \sin\frac{n\pi x}{l}$$

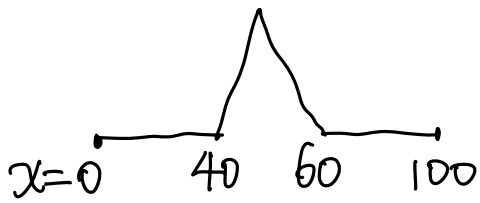
$$\Rightarrow C_n \frac{n\pi a}{l} = \frac{2}{l}\int_0^l \psi(x) \sin\frac{n\pi}{l} x \, dx$$

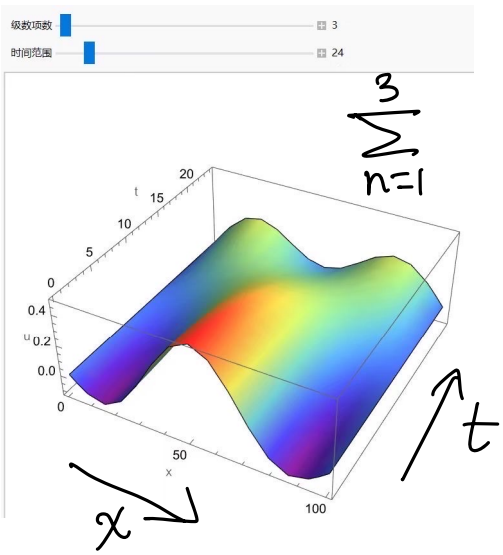
$$\Rightarrow C_n = \frac{2}{n\pi a}\int_0^l \psi(x) \sin\frac{n\pi x}{l} dx$$

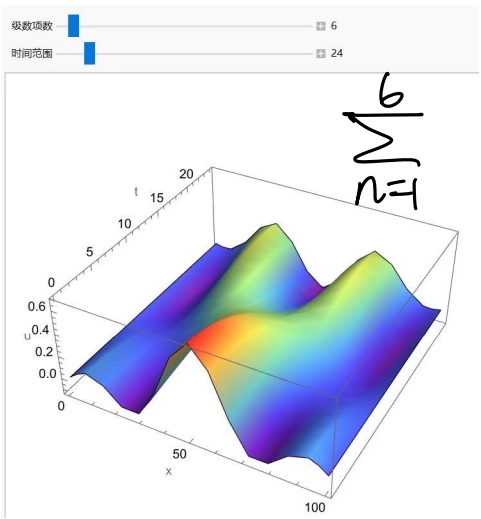
图像展示. 取 $a=1$, $l=100$

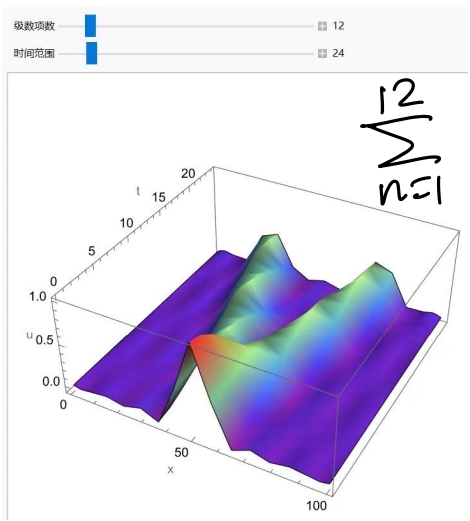
波动方程 $u_{tt}-u_{xx}=0$，初始速度 $=0$，初始位移为 tent function

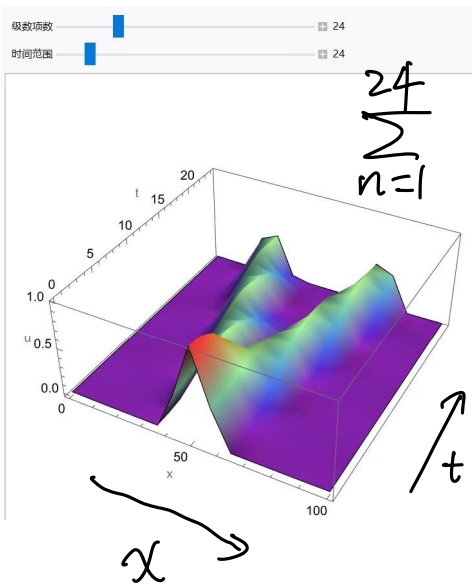
帐篷函数











较少项数即可呈现基本轮廓，较大几补足细节特征

物理图象：

$u_n(x,t) = X_n T_n = \sqrt{C_n^2 + D_n^2}\sin\left(\frac{n\pi a t}{l} + \theta_n\right)\sin\frac{n\pi}{l}x$ 驻波

不同频率的驻波 $\xrightarrow{\text{叠加}}$ 解 ← 满足初始条件

↑ 一般不满足初始条件

驻波解与行波解殊途同归。驻波可改写为右行波加左行波：

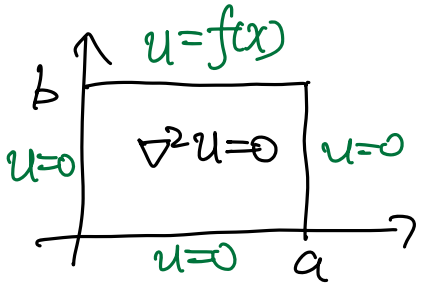
例：$\sin\frac{n\pi a}{l}t\sin\frac{n\pi x}{l} = \frac{1}{2}\left[\cos\frac{n\pi}{l}(x-at) - \cos\frac{n\pi}{l}(x+at)\right]$

驻波 ↑ 右行波 ↑ 左行波

其他依同理.

Laplace 方程：$\nabla^2 u = 0$ $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$

$u|_{x=0} = u|_{x=a} = u|_{y=0} = 0$, $u|_{y=b} = f(x)$



(x, y) ⟷ (x, t)

Laplace　　波动

作为作业